

Figure 1: Ackley's function

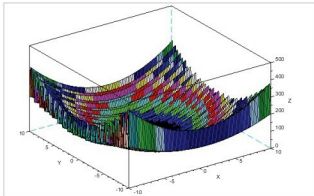


Figure 2: Levi function

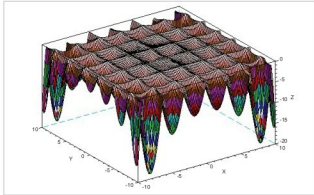


Figure 3: Holder table function

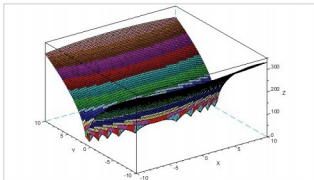


Figure 4: Buckin function

$$z = -20 \times e^{-0.2 \times \sqrt{0.5 \times (x^2 + y^2)}} - e^{0.5 \times [\cos(2\pi x) + \cos(2\pi y)]} + e + 20 \quad (1)$$

$$z = \sin^2(3\pi x) + (x-1)^2 \times (1 + \sin^2(3\pi y)) + (y-1)^2 \times (1 + \sin^2(2\pi y)) \quad (2)$$

$$z = -|\sin(x) \times \cos(y) \times e^{1 - \sqrt{(x^2 + y^2)/\pi}}| \quad (3)$$

$$z = 100 \times \sqrt{|y - 0.01 \times x^2| + 0.01 \times |x + 10|} \quad (4)$$

Figure 5: Toy functions

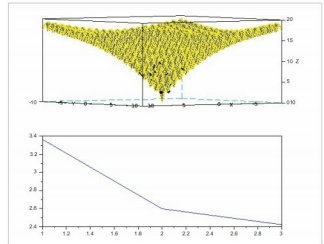


Figure 6: Function convergence, initial stage

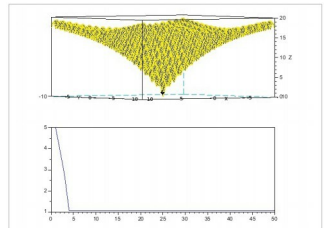


Figure 7: Final function convergence

for convergence using contour plots and other performances can be obtained with a few tweaks within the algorithm. For now, we have discussed the general idea of implementing toy function optimisation in Scilab. 

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